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Cross Industry Analytics Solution Library for Resource and Operations Management

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CROSS INDUSTRY ANALYTICS SOLUTION LIBRARY FOR RESOURCE AND OPERATIONS MANAGEMENT

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ABSTRACT

Inthispaper,wedescribeaframeworkofanalyticssolutionlibraryforsolvingvarietyofindustrial problemsinresourceandoperationsmanagement.ThisBigDataenabledanalyticssolutionslibraryis targetedformanufacturingindustriesincludingElectronics,Semiconductor,Automotive,Chemical& Petroleum, Oil & Gas industries, Energy & Utility, and Mining & Metals industries.The analytic library contains families of state of art analytics including predictive asset management, predictive failure analysis ofprocessandequipment,processandequipmentmonitoring,analysisandoptimization,predictive environmental analytics system and spatio-temporal analytics for safety and operational effectiveness.The analytics is powered by variety of advanced algorithms, is cataloged based on a comprehensive ontology, and guides common users without specialized training in mathematics with application specific workflows and a pre-engineered application programming interface (API), data structures and Graphical User Interface (GUI)widgetsforvisualizinginputdataandoutputsolution.Theanalyticssolutionlibrarycaningest remote sensing data from the Internet of Things (IoT) 3.0 foundations and is designed to consume variety andvastamount(i.e.,BigData)ofstructuredandunstructureddataincludingvideo,textandrealtime sensor data.The analytics services library bridges the gap between analytics algorithms and real business problems, and promotes the convergence of Information Technology (IT) and Operations Technology (OT) andpromotesavisionofIndustry4.0througheffectiveandefficientutilizationofadvancedanalytics provisioned through an increasingly sensored physical world.

1 INTRODUCTION

Aneweraofcomputingishere. Theneweraofcomputingistheeraofcognitivecomputingthat understandstheworldinthewaythathumansdothroughsensing,learning,andadaptingbehavior automatically based on new knowledge.The first computing era started around the 1900s and was made upoftabulatingmachinesandtheseconderastartedaround1950sandwasmadeupofprogrammable computers.Cognitive computing represent a whole new approach to solving complex problems with large amount data, machine learning and analytics that goes beyond just computing.There are enormous growth ofBigData,whichisbeingfueledbytheconfluenceoftheinternetofthings(IoT),social,mobileand cloud.Industrial operations, processes and equipment are now also generating enormous amount of data. Currently, one trillion connected objects and devices on the planet are generating 2.5 billion gigabytes of data every day, and the rate is growing.The size of the machine generated data globally is about 8.5 ZB (zettabytes, or trillions of gigabytes), and is expected to grow to exponentially to 40 ZB in 5 years (2020) [1].By 2020, machine to machine data will dominate constituting 40% of all the data.The data generated from sensors and devices are becoming bigger than the social media data, VoIP and enterprise data.

Thedataarebeinggeneratedfrom varietyofplacesandobjectsincludingmanufacturing shopfloor, industrialequipment,sensorsfromfactoriesandhomes,socialmedia,smartphone,satelliteimages, weather laboratory, home appliances and even from automobiles that we drive.The types of the data are also very diverse.Data can be structures, un-structured, video, audio, text and so on.

Data has become the world's new natural resource.Industry is being revolutionized with the data.By utilizingthedataproperly,manufacturingfirmscanimproveproductquality,reduceproductioncosts, improvesafety in manufacturing plants, improveenergy efficiency ofproduction,and improvethe profit margin.Data is now the basis of competitive advantage.The data can also be used improve the quality of human lives by promoting smarter buildings, smarter cities and smarter planet.However, collection of the Big Data itself does not generate any benefits.It is important to interconnect all the data sources, and to collect the big data; however, without appropriate analytics to process the data and transform the data into useful decision supporting information, the Big Data can be just a big waste.Only when the data is properly analyzed and utilized, the data can help industries and society.Analytics in this new era of computing is different from conventional analytics.The traditional analytics can no longer be able to handle the volume, variety, velocity and veracity of Big Data.What we need in the new era of computing is advanced analytics that can be easily built, configured and deployed, and fully integrated with IoT and Big Data.The analytics intheneweraofcomputingiscomplexaggregationofvariousalgorithmsandtechniquesincluding advanced mathematics, statistics, data mining, machine learning, cognitive computing and optimization.

One important technical domain in industry that can take advantage of the analytics in the new era of computing is *Resource and Operations Management*, which is a collection of key problems that deals with assets, resources,operationsandprocesses in manufacturing industries. The ability to integrate withIoT, andstore,processandanalyzeBigDatawillimprovevariousaspectsofresourceandoperations management problems, and bring competitive advantage to industries.

By effectively and efficiently utilizing Big Data andIoTwith analytics, the opportunities to improve thevariousindustrialprocessesandoperationscanbediscovered.However,itisquitechallengingto develop such analytics because it typically requiresdeep expertise across datamining,machine learning, statistics and optimization as well as domain knowledge in order to develop practical solutions. Substantial resources and time are required to develop such analytics.Therefore, not many enterprises can afford to developandutilize such analytics. Itisoftendifficultfor commonusers withoutspecializedtraining in mathematicalmodelingorstatisticstoidentifyanappropriateanalyticsforsolvingaspecificproblem, developandruntheanalyticsinappropriatestepsandmannerstoproducedesiredresults.Increasingly alongwiththecomplexityofthedataitself,thecomplexityofchoosingtherightanalyticsalgorithmsis emerging as another challenge in the IoT era.

In this work, we are developing a solution platform that provides an analytics services library, called SROM (Smarter Resource and Operations Management), with a hierarchy of analytics that can solve variety of problems in the area of resource and operations management.The platform enables the common users without specialized training in mathematical modeling to easily identify, develop and use the analytics for solvingcomplexindustrialproblemsinresourceandoperationsmanagementwithmuchlesseffortand time.

2 CAPABILITIES OF SROM

Smarter Resource and Operations Management (SROM) is a next generation analytics solution framework that can fill the gap between real business and analytics in the areas of resource and operations management. Most of the existing analytics solutions for industrial problem lack realistic use casesand industry specific solutions.Asignificantgapexistsbetweenalgorithmsandrealindustrialproblems.SROMintegrates business solutionlibrary thatisprovisionedby rich,state-of-the-artalgorithms.Analyticsplatformsfor enterprisesystemsarebeginningtoevolvefrompureanalytics'functionenginestobusiness/enterprise functionorientedsolutionsthatarepurposebuiltforspecificindustries.Thefirstgenerationanalytics

typicallyconsistsofbarestatisticsfunctionandmachinelearninglibrary,andtheexamplesofthese analytics include products such as S, R, Weka, etc. The second generation analytics are typically machine learning library armed with composable GUI enterprise system, and the examples of these are SAS, SPSS, etc.The third generation analytics are cloud-based, solution-oriented, function based-APIs, and SROM is one of them.SROM is designed to address the rapidly evolving solution needs for asset heavy industries.

SROM is an ontology based services solution library, and it is enabled to be integrated with Big Data and IoT.SROM is designed for cross-industry reuse, and targeted for manufacturing industries including Electronics, Semiconductor, Automotive, Chemical & Petroleum, Oil & Gas industries, Energy & Utility, Mining & Metals industries.The analytics are powered by variety of advanced algorithms, are cataloged basedoncomprehensiveontology,andguidecommonusersthoughapplicationspecificworkflowwith pre-engineered application programming interface (API), data structures and GUI widgets for visualizing input data and output solution.

The analytic library includes, but not limited to, families of state of art analytics such as predictive asset management, predictive project health, process monitoring and optimization, and spatio-temporal analytics for safety and operational effectiveness as can be seen in Figure 1.SROM is a cloud-enabled services, and it is integrated with data services, which have connected to two types of data.One type of data is IoT data, which is dynamically collected from sensors and equipment in industrial operations.This data can include data from SCADA (supervisory control and data acquisition) systems, DCS (Distributed Control System), MES (Manufacturing Execution System) and other OT system as well as IT system etc. The other type of data is systems of record such as ERP (enterprise resource planning) system and work order management system and asset management system and so on.SROM analytics consume both types of data.Depending ontheparticularproblem ofinterest,thedatasizecanbeverylarge(i.e.,BigData),andthedatacanbe structured and unstructured data including video, text and real time sensor data.SROM analytics services layer in the middle of Figure 1 can be packaged into specific industry specific solution pack, for example, an analytic library packaged for automotive industry, chemical/ petroleum, electronics or energy/utility etc.

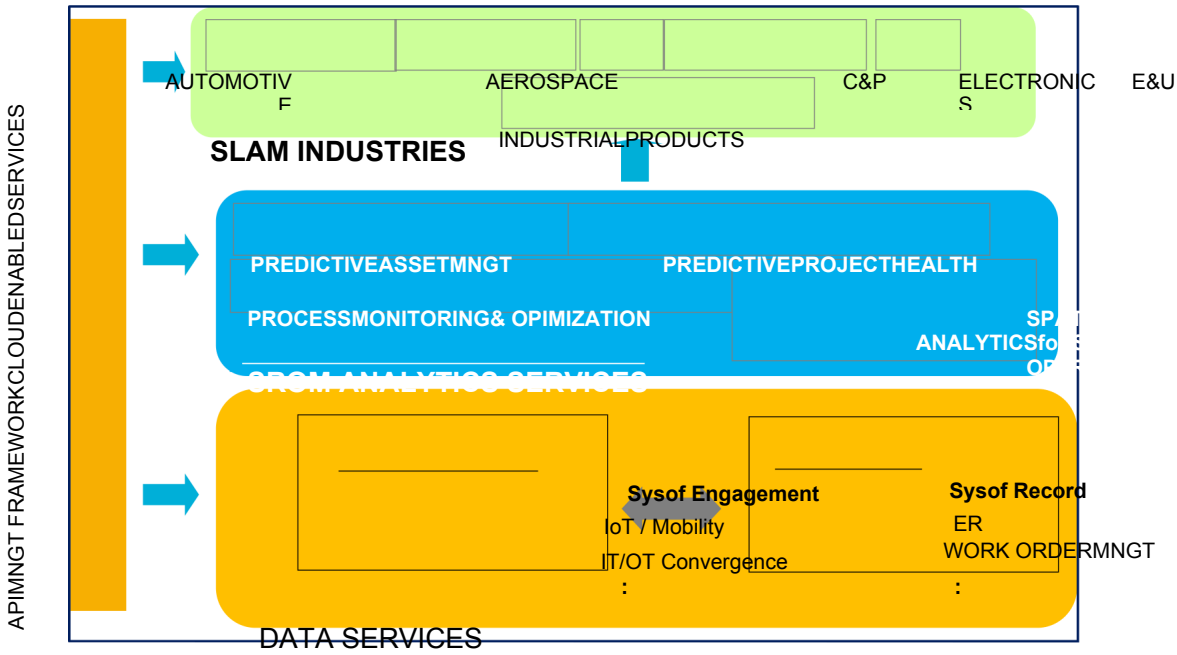

Figure 1: An Overview of SROM

Therearemanytypesofanalyticsthatcansolvevariousproblemsinindustryintheareasof resourceandoperationsmanagement,butitisoftendifficultforcommonuserstoidentifyappropriate analytics and algorithm for solving a specific problem, and also to try out example applications or use cases that are already built for other firms in the same industries. The analytics in SROM is catalogued based on ontology of analytics.The ontology enables the common users without specialized training in mathematical modelingtounderstandwhataretypicalanalyticsthatareusefulindifferentindustries,howvarious analytics are related to one another, identify, and ultimately develop a specific analytics for solving complex industrialproblemsin resource and operations managementwithmuchless effortand time.A simplified view of the overall structure of SROM ontology is shown in Figure 2.

As shown in Figure 2, there are four levels of elements in SROM ontology in the area of resource and operations management.The top level is the family of analytics and they includes: *Predictive Maintenance*, *Predictive Failure Analysis*, *Process and Equipment Analysis* and *Process Monitoring and Optimization*.

Predictive MaintenanceAnalytics is used to determine the condition of various assets and equipment inprocesses and operations and predict when maintenance should be performed.Themaintenance work can be inspection, repair or replacement of equipment and parts.This analytics are typically formulated as optimization model such as Mixed Integer Linear Programming (MILP) with a goal of minimizing objective functionofcostofmaintenanceandfailureandwithconstraintsofresources,spareparts,budgetand operations. Predictivemaintenanceanalyticshelptoreducemaintenancecost,increaselifetimeof equipment and increase the quality of products and improve the safety of manufacturing plant.A longer termplanningactivityofPredictiveMaintenanceisMaintenancePlanning,whichdeterminesa maintenance plantypically in weekly or monthly time bucket with a planning horizon of one or more years consideringfailureriskofasset. AshortertermschedulingactivityofPredictiveMaintenanceis MaintenanceScheduling,anditdeterminesschedulingofmaintenancecrew,resourcesandrouting typically in hourly or sub-hourly time resolution for a time horizon of one day or one week.

Predictive Failure Analysis is used to predict future failure of assets and equipment so that corrective actions are taken to avoid the failure.Failure of assets and equipment in production can cause significant negative impact on product quality, production cost and safety.There are two types of predictive failure analysis.OneisFailurePatternAnalysis,whichidentifiesconditions(alsoreferredaspatterns)thatare precursorstoafailure.TheotherisFailureRiskAnalysis,whichestimatesofthelifeexpectancyofan asset based on historical failure data and operational conditions.

Process and Equipment Analysis is used to analyze the stream of sensor data generated from the process and equipment and identify anomalies or faults.The analysis is also called Fault Detection and Diagnosis (FDD).AgoodinstanceofthisanalysisisAnomalyDetection,whichidentifiesassetbehaviorthatis outsideofadefinednormthatconstitutesnormalbehavior. Anomalydetectionfirstcharacterizesthe normal behavior using historical measurements of the asset behavior under different conditions, and then identifies the excursions outside these norms as potential anomalies, and generates alerts.

Process Monitoring and Optimization refers to a group of analytics that continuously monitors process and equipment, and optimizes the controllable variable settings so that the processes or operations run in most cost-efficient, energy-efficient and safe manners with maximized throughput.A predictive model for the processes or operations, either in physics model or data-driven model or combination of both needs to be built for this analysis.And then an optimization model is built to compute the optimal control variables values using on the predictions from the predictive model.

Foreachanalyticsfamily,therearemultipleinstancesofanalytics(secondlevel).Forexample,the *Predictive Maintenance*has *Maintenance Planning* and *Maintenance Scheduling* analytics among others. And, *Predictive Failure Analysis* family contains the analytics of *Failure Pattern Analysis* and *Failure Risk Analysis*.Eachanalyticscanbesolvedbymultiplealgorithms(level3).Forexample,*FailurePattern Analysis*canbesolvedbythreedifferentalgorithms,i.e.,*DecisionTree*,*SupportVectorMachine*and *Random Forest*.Then at the last level (level 4), sample applications of use cases are included.For example, for*GraphicalMethod*of*AnomalyDetection*analyticshasusecasesof*MiningMachineryAnomaly Detection*and*OffshoreOilPlatformAnomalyDetection*.Themoredetailedhierarchiesof*Predictive*

*Maintenance*and*PredictiveFailureAnalysis*areshowninFigure3.Notethattheparticularontology shown here is still being developed and are not complete.Continuous work is being done to streamline and improve the ontology as we deal with more industrial problems.Some instances of the analytics, algorithms and use cases are described in the section 6 below.

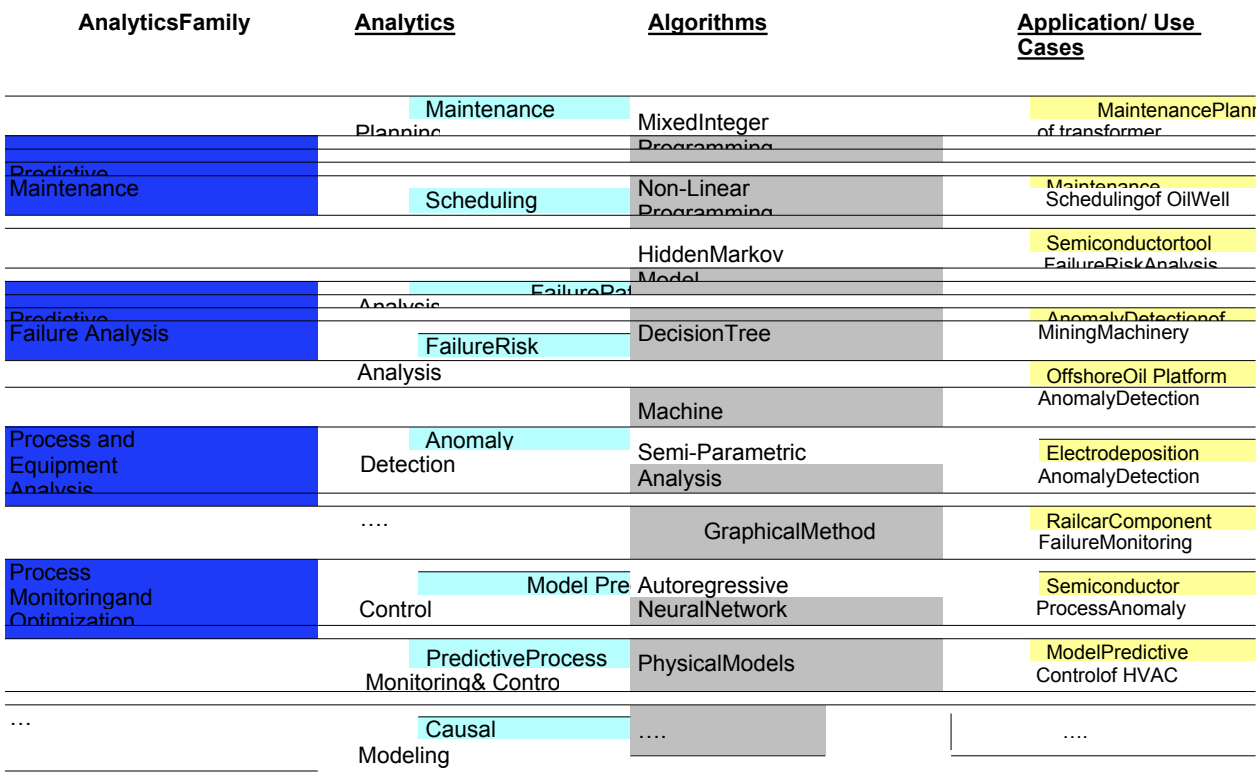

Figure 2: An Overview of SROM Ontology

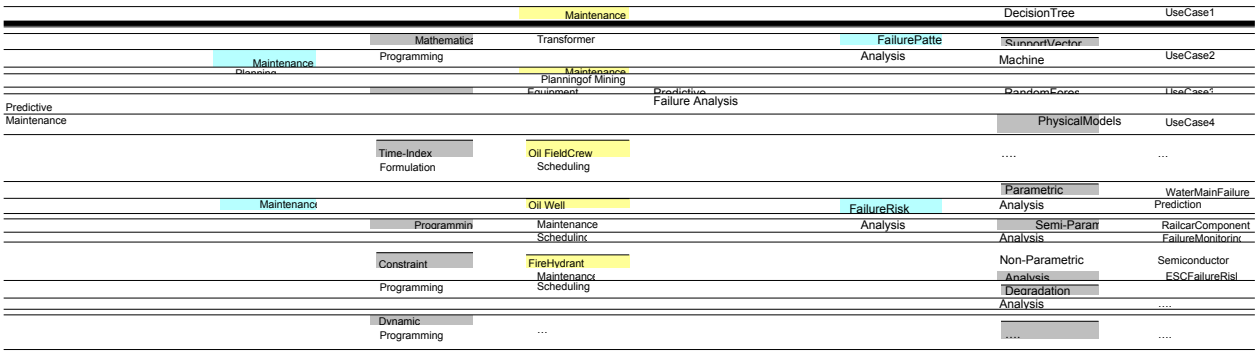

Figure 3: Ontology of Predictive Maintenance and Predictive Failure Analysis

Each analytics uses specific input data and produces certain output data as solutions.In addition, the solution of one analytics can be used as an input to another analytics.An example of the relationship among analyticsandcorrespondingdataisshowninFigure4.Inthisexample,dataoffailurehistory,asset attributes, environmental data and operational data are used for *Failure Pattern Analysis* model and *Failure*

*RiskAnalysis*model.Theoutputfrom the*FailureRiskAnalysis*model,hazardfunctionalong withcost factorsofmaintenanceandfailure,reliabilityestimationandoperationalconstraintssuchasbudgetand resourceofmaintenance,areusedasinputfor*MaintenancePlanning*optimizationmodel. The maintenance plan and anomaly detected can be used as input to *Maintenance Scheduling* model.

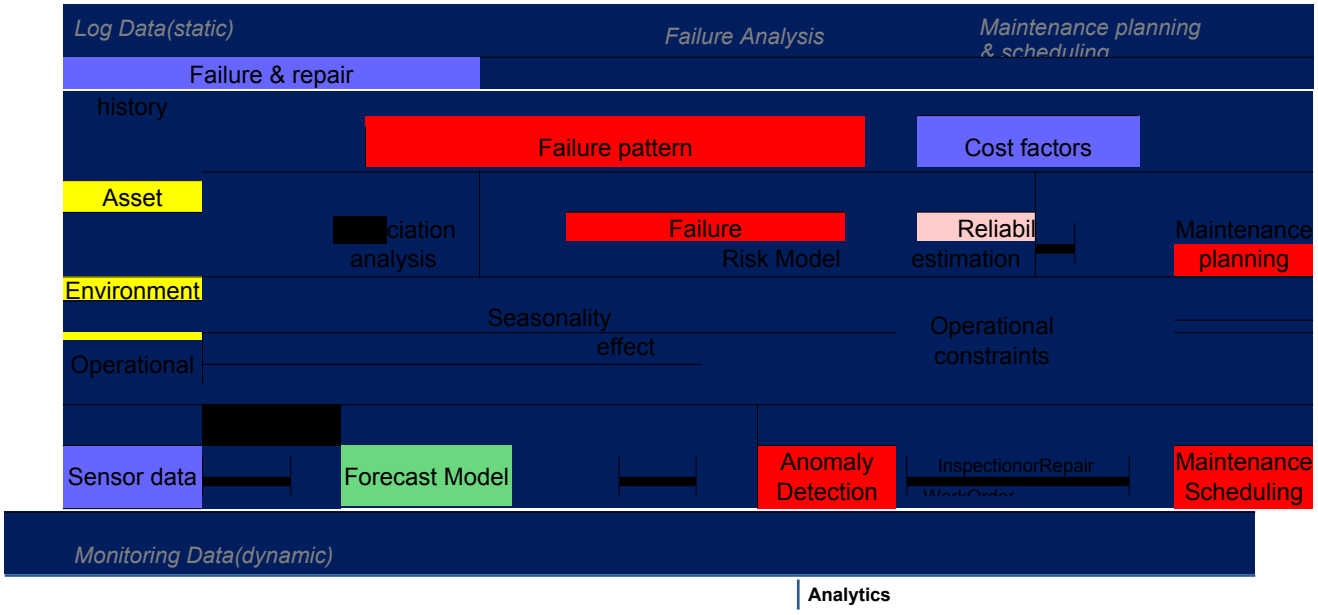

Figure 4: Analytics and Data Relationship

The ontology of analytics is implemented as a hierarchical menu for easy navigations in a web-based GUI as shown in the left side of Figure 5.

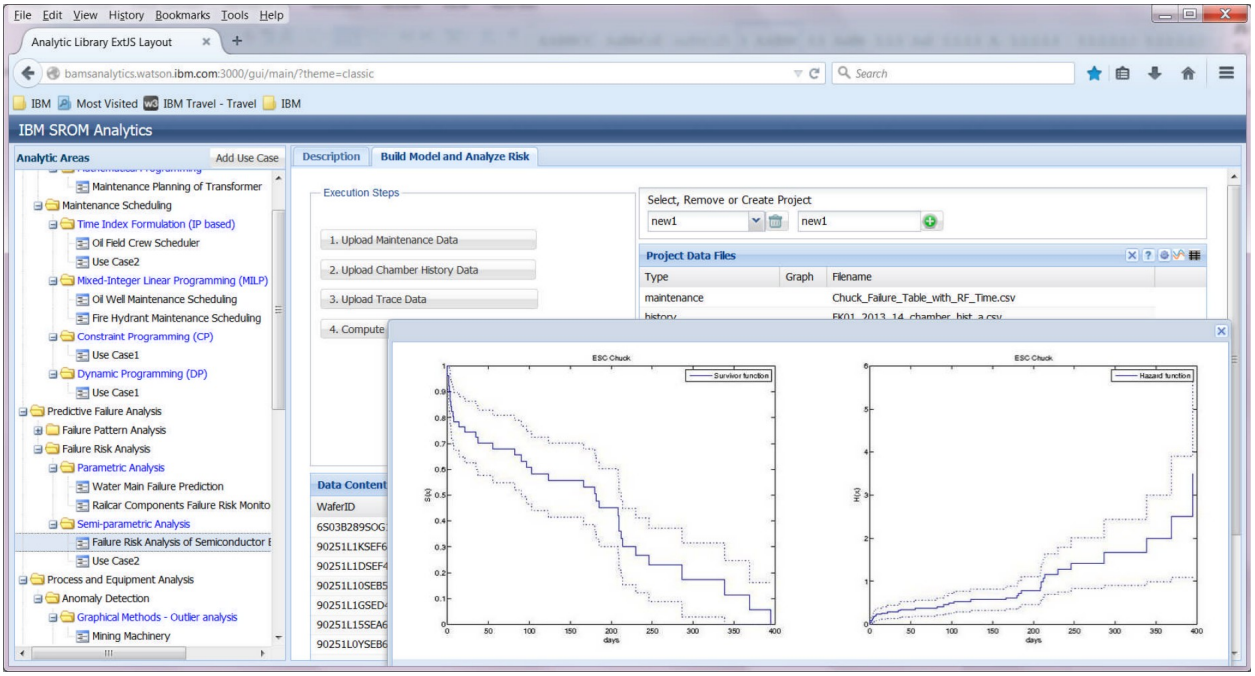

Figure 5: An Example for the Semantically defined SROM Analytics Solution

3 ARCHITECTURE OF SROM

The core of the SROM is the advanced analytics library that consists of analytics families such as *Predictive AssetManagement*,*PredictiveProjectAnalytics*and*ProcessMonitoring/Optimization*etc.asshownin Figure 6.A subset of this library can be packaged for a specific industry as an industry pack.For example, *industry pack – C&P* is a subset of SROM library for chemical and petroleum industry and *industry pack –E&U*isforenergyandutilityindustry.Theanalyticslibraryissemanticandontologybasedsothat common users can easily identify a specific analytics for a specific problem to solve by navigating through the hierarchy of the ontology.

The SROM analytics are supported by a computational layer, where computational engines are located. Fortheanalyticsthatinvolveoptimizationanddecisionmanagement,anoptimizationenginesuchas ILOGTMCplex, an optimization platform such as ILOG Decision Optimization Center (DOC) and decision management tool such as SPSSTMDM can be utilized.For the analytics that involve statistical analyses, statisticalenginessuchasSPSSModeler,R,PythonorMatlabcanbeaccessed.Fortheanalyticsthat involvebusinessanalyses,abusinessintelligencetoolsuchasCOGNOSTMBI canbeaccessed.Forthe analytics that require visualization and discovery, Watson Explorer can be accessed for example.

Thecomputationalenginesconsumevarietyandvastamount(i.e.,BigData)ofstructuredand unstructured data including video, text and real time sensor data, and this is supported by data storage layer. Thedatastoragelayerincludesdatarepositorysuchas HDFS,HBase,Hive,Sparkforarchivinglarge volumesofoperationaldataandsensordata.Italsoincludesdatabasesystemsuchas Cassandra,DB2, Informix,PureDataAnalyticswithpre-builtdataschemaforstoringquality,machineandproddata, configurati For IoT applicatio real-time event processi capabil such as InfoS phere Streams can also be utilized.

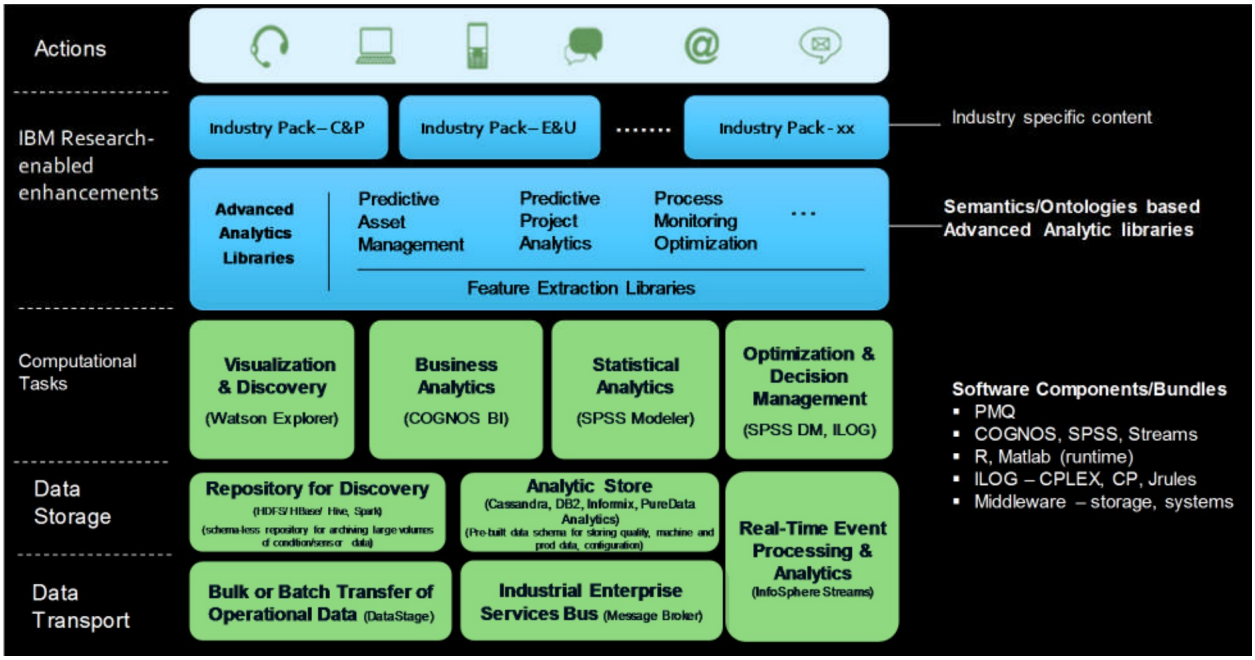

Figure 6: An Architecture for the Semantically defined SROM Analytics Solution

Thedatastoragelayerissupportedbydatatransportlayerwhichenablesbulkorbatchtransferof operational data for data stage, and industrial enterprise services bus, i.e., message broker.

SROMisacloud-basedsolution,andthesolutioncomputedbytheSROManalyticscanbe communicated to users through computer screen, mobile devices, email, social media and voice etc. in the action layer.

4 SROM WITH IOT AND BIG DATA

The Internet of Things (IoT) is a technological revolution in the future of computing and communication that is based on the concept of anytime, anyplace connectivity for anything [5].IoT represents a network of Internet-enabled objects and devices, such as factory equipment, consumer electronics, home appliances, sensors, meters and smart phones. We are already using IoT solutions in practical ways in our daily lives. For example, we can monitor our home security, lights, heaters, air conditioners and status of other electric appliances from our smartphones.In manufacturing environment, even more data are generated in faster speed from various SCADA (supervisory control and data acquisition) systems, DCS (Distributed Control System),MES(ManufacturingExecutionSystem)andotherOTsystemaswellasITsystemetc.IoT affectsalmostallindustriesincludingautomotive,chemicalandpetroleum,miningandmetaland electronic.

Big Data is a phenomenon that is characterized by rapid accumulation of data from numerous sources. Thedataaregeneratedandcollectedsoquicklyinsuchawaythatitisfloodingindustriesandsociety. Thedataarevaluable,newrawmaterialsinindustries.Wecanutilizeittoproducenewservicesand products,andtoimprovetheexistingservicesandproducts. Therefore,BigDatarepresentsbotha challengeandanopportunity.Thechallengeisrelatedtohowthisvolumeofdataisharnessed,andthe opportunity is related to how the effectiveness of industries and society is enhanced by properly analyzing and utilizing this information.It is now commonplace to distinguish Big Data solutions from conventional IT solutions by considering the following four dimensions [3]:

- Volume. Big data solutions must manage and process larger amounts of data.
- Velocity. Big data solutions must process more rapidly arriving data.
- Variety. Big data solutions must deal with more kinds of data, both structured and unstructured.
- Veracity. Big data solutions must validate the correctness of the large amount of rapidly

BM quantum computers leverage furniture data.

Big data is being generated by everything around us at all times. Every digital process and social media exchange produces it. Systems, sensors and mobile devices transmit it. Big data is arriving from multiple sources at an alarming velocity, volume, variety and veracity.

Big data and the IoT are computing paradigms that, together, fundamentally change the nature of how wework,play,andinteractwithourenvironment.WhereBigDataisallaboutvolume,velocity,verity, and veracity, the IoT is about using that data in meaningful ways to improve productivity and quality of life [6].However, IoT and Big Data by themselves are not useful.Only when the data are properly analyzed andconvertedtousefulinformation,benefitsofBigDatacanberealized.Analyticsarethekeyfor extractingmeaningfulvaluefromBigData. BigDataischangingthewaypeoplewithinorganizations worktogether.ItiscreatingacultureinwhichbusinessandITleadersmustjoinforcestorealizevalue fromalldata.Insightsfrombigdatacanenableallemployeestomakebetterdecisions—deepening customer engagement, improving product quality, optimizing operations, preventing threats and fraud, and capitalizing on new sources of revenue [2].But all these are possible only through effective use of analytics. Analytics is the collection of advanced mathematics, statistics, data mining, machine learning and cognitive computing.The intent of the Cross Industry Analytics Solution Library, e.g., SROM is to rapidly distribute the analytics capability throughout industries.

SROM (Smarter Resource and Operations Management) is a library and a development platform that containsasetofanalyticsthatcansolvevarietyofproblemsintheareaofresourceandoperations management. Theplatformenablesthecommonuserswithoutspecializedtraininginmathematical modelingtoeasilyidentify,developandusetheanalyticsforsolvingcomplexindustrialproblemsin resource and operations management with much less effort and time.

The description of SROM's platform with IoT and Bid Data are shown in Figure 7.The source of data for SROM includes smart devices such as smart phones and tablets, IT and OT systems from manufacturing plants such SCADA system, and other external sources for weather data, geographical data and traffic data etc. as shown in the bottom of Figure 7.These IoT data are serviced by IoT operation service layer, which includesIoTDataInfrastructureServices,IoTServicesCompositionandLifecycleServices,IoTDevice Connectivity/ManagementServices,IoTPlatformOperationAnalyticsServicesandIoTSecurity IntelligenceServices.ThedatacollectedfromIoTlayerareaggregatedanddistributedintheBigData layer,whichincludeHadoopDistributedFileSystem (HDFS),SparkandMapReduceetc.TheBigData layersisaccessedby SROManalyticslayers,whichincludethefamiliesofvariousanalytics inresource andoperationsmanagement.SROManalyticservicesarepackagedintovariousindustrysolutionpack such as mining and metals, chemical and petroleum, and electronics etc.

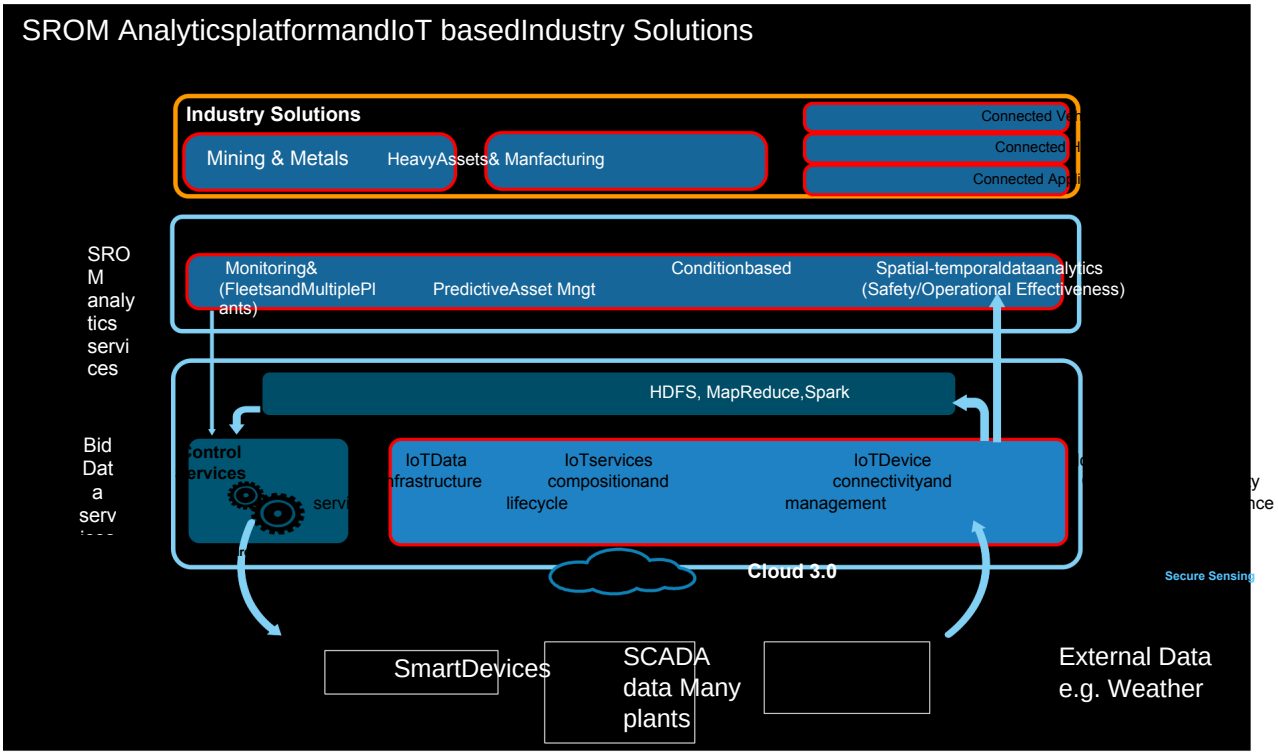

Figure 7: SROM Analytics platform and IoT/Big Data based Industry Solutions

5 DEPLOYMENT OF SROM

The analytics in SROM require substantially configured computing hardware and software including data interfaceprograms,database,dataintegrationandtransformation,datawarehouse,statisticalengine, optimization engine, and visualization.Also depending upon the size of the data and amount of analysis thereisaneedtoscaleupandscaledowntheinstallbase.Typicalusersmaynotaffordsuchresources individually.The cloud is a secure, flexible and scalable way of delivering such analytic services to many users by sharing all the resources in a cost effective and secure way among many users.Running this kind of analytics service on cloud for multiple clients using the shared computing resource is more cost-efficient thanprovidingthesameservicetoindividualclientswithatraditionalcomputingmediumwithoutthe cloud.Dependingonthenumberandrolesoftheusers,anappropriatelevelofresourcecanbeeasily configured virtually on the cloud, and the level of access to resources can be adjusted on demand.Using **cloud, all the analytics can be redaily accessible by the users through web browsers without installing any** special hardware, software, data interface and mathematical tools.

SROM is designed to be deployed as Analytics as a Service (AaaS) model.SROM is hosted as a service and is provided to customers across the Internet.Deployment of SROM on the cloud eliminates the need to install and run the analytics on the customer's own computers along with alleviating any burden to the customer with regard to software maintenance, ongoing operation and support.SROM cloud deployment is designed to provide rapid access to security-rich, enterprise-class virtual server environments, well suited for dynamic workloads.SROM cloud offers the capabilities to control access and configure security.

6 DESCRIPTIONS OF ANALYTICS IN SROM

A few selected analytics are described here.SROM is still under development and additional analytics and use cases are being added.

6.1 Maintenance Planning

Maintenance planning is targeted at providing a cost minimizing maintenance plan over the lifetime of an asset. Thisanalysisconsidersthecostof**erpair,**downtimeandreplacementtoprovidearepair/replace plan.

The planning horizon is the lifetime of the asset and hence for high capital assets it could be decades. Themaintenanceplanningisanoptimizationexercisethatprescribestheoptimaltimeformaintenance, repair and replacement.The planning exercise requires as input the failure risk analysis (i.e. the survival function) which provides the life expectancy of an asset based on its history and conditions of use.

Use Case: Maintenance Planning of Transformer

This optimization application computes an optimal preventive maintenance/replacement schedule for a set of assets under given constraints such as budget and labour. The input data for this models includes asset information, hazard function data and constraints such as budget and personnel availability for maintenance. Thisusecasecomputestheoptimalmaintenanceplanningoftransformersofanutilitycompany.The optimalproblem isformulatedand solvedasNon-LinearProgramming (NLP)andMixedIntegerLinear Programming (MILP).

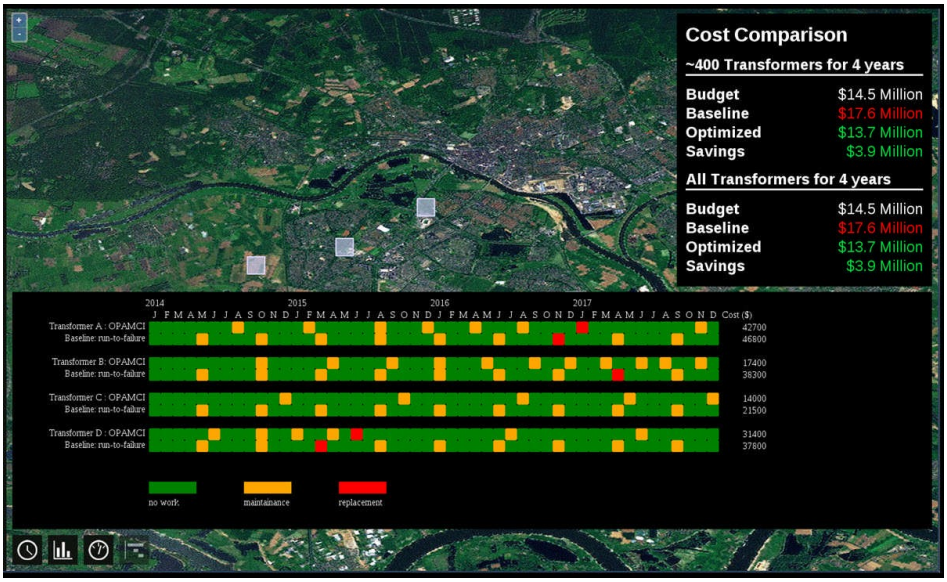

Figure 8: A Sample Output of a Maintenance Plannign Optimization Analytics

6.2 Maintenance Scheduling

Maintenance scheduling is an operational scheduling effort with a horizon of weeks or months.Consider a plant with a portfolio of assets.Each asset has a maintenance plan over its life time.Now for any given **month (or week) each asset can be examined to see if there is a prescirbed inspection/repair work order.**If wecollectalltheactivitiesthatneedtobeperformedwithinthemonththenwehaveaproblemof scheduling the work orders keeping in mind constraints re crews, vehicles, spare parts inventory and tools. Thisschedulingmightalsorequireconsiderationsofthedifferentlocationsandtraveldistances.The scheduling exercise is a prescriptive analysis that takes as input the work orders, the required resources, the

timewindowswhen thework orderneedstobeperformedandgeneratesa cost minimizing schedulefor maintenance.

Use Case: Maintenance Scheduling of Oil Wells

Thisapplicationcomputesoptimizedschedulingandroutingofmaintenanceresources(e.g.operators, equipment, material etc.) for maintenance and inspection of oil wells so that maintenance requests can be fulfilledatminimalcost.Thetoolallowsuserstorouteandschedulepeopleandresourcestoattend geographically sparse tasks.The user inputs the locations and durations of tasks, which may be different depending on the types of technicians and equipment which are used to complete the task. The tool then uses Mixed Integer Linear Programming (MILP) techniques to search for a solution which assigns as many tasks as possible within a given time frame, whilst minimizing travel distance and total time spent.

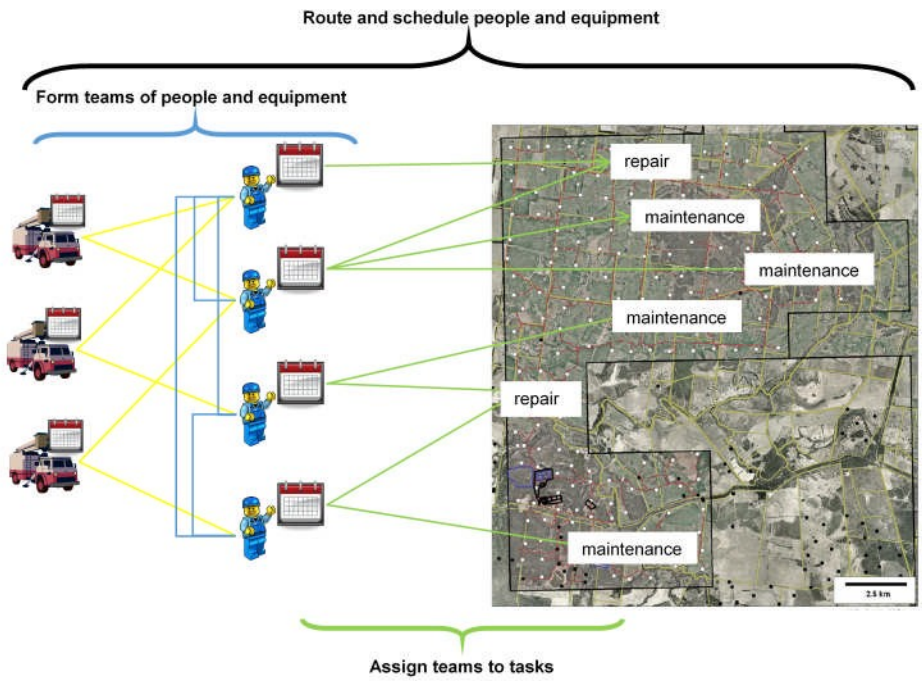

Figure 9: Overview of a Maintenance Scheduling Optimization Analytics

6.3 Failure Risk Analysis

Failure Risk Analysis estimates the life expectancy of an asset based on historical failure data for assets of thatclass.Thelifeexpectancyofanassetdependsonitslifehistory,whichischaracterizedbyitsown attributesand the environmentalandoperationalstress factors.A typical failure risk modelprovides the survivalrateasafunctionofthevariousenvironmental,operationalfactorsand itsownattributes. The survivalrateprovidestheprobabilitythatassetwillsurviveagivendurationintothefuturebasedonits ownhistory.Survivalanalysisconsistsofparametric,semiparametric,andnonparametricmethods.Itis usedtoestimatethemostcommonlyusedmeasuresinsurvivalstudies,survivorandhazardfunctions, compare them for different groups, and assess the relationship of predictor variables to survival time.

Use Case: Failure Risk Analysis of Semiconductor Electrostatic Chuck (ESC)

This application predicts the life expectancy of Electrostatic Chuck (ESC) in semiconductor manufacturing process.Based on historical maintenance data, operational history and sensor data from the manufacturing equipment, this tool predicts survivor/failure function, and identify what/how process variables impact the chuckfailure,andidentifytheconditionsleadingtoESCmaintenanceortoESCreplacement.Therisk analysis is then used to compute the optimal maintenance and replacement plan that minimize the cost of maintenanceandfailure. ThisanalyticsisbasedonCoxproportionalhazardsregression,whichisa semiparametricmethodforadjustingsurvivalrateestimatestoquantifytheeffectofpredictorvariables. Themethodrepresentstheeffectsofexplanatoryvariablesasamultiplierofacommonbaselinehazard function. ThehazardfunctionisthenonparametricpartoftheCoxproportionalhazardsregression function, whereas the impact of the predictor variables is a log linear regression.

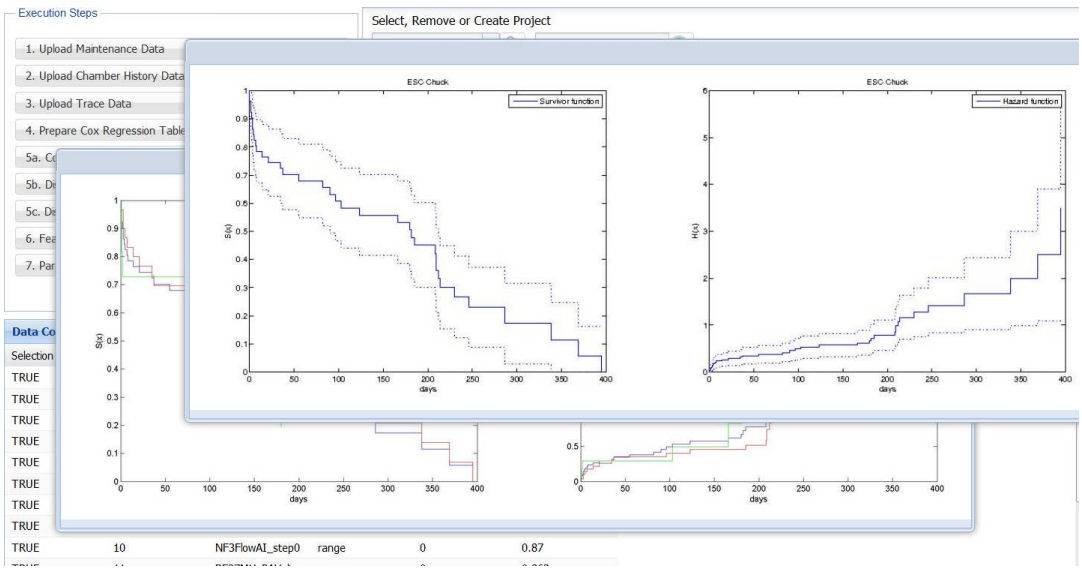

Figure 10: A Sample Output from a Failure Risk Analytics

6.4 Anomaly Detection

Anomaly Detection is the task to detect behaviors that look differently from normal operations. Typically anomaly detectionneeds three components: (1) A normalstatemodel.Inmodernanalytics,a probability distribution is used to model the normal state for practical application. (2) A definition of anomaly score, which is generally defined as a function of the probability distribution of the normal state. (3) A threshold to the anomaly score to decide whether or not the current observation or state is anomalous. Among several subtasksthatcanbedefinedbasedonanomalydetection,changedetectionisofparticularpractical importance for IoT sensors. In change detection, the sliding window techniques is typically used to define the past and the current states, and a score representing the excursion of the current state from the past is computed.

<u>Use Case: Mining Machinery Anomaly Detection</u>

The application detect anomalies from expensive mining machineries such as shearers in underground coal mining.In this mining operations, sensor data obtained from mining machineries are highly dynamic and traditional outlier detection approaches do not work. Due to the intractable nature, the monitoring solution

had suffered from toomany false alerts. To tackle the challenge, this use case leverages the technique of sparse structure learning and formalizes the anomaly detection task as quantification of structural changes of inter-variable dependencies, as represented by Figure 11.Here the highest anomaly score is attributed to a major structural change with respect to the 24th variable, whose dependency with 40th variable does not exist any more

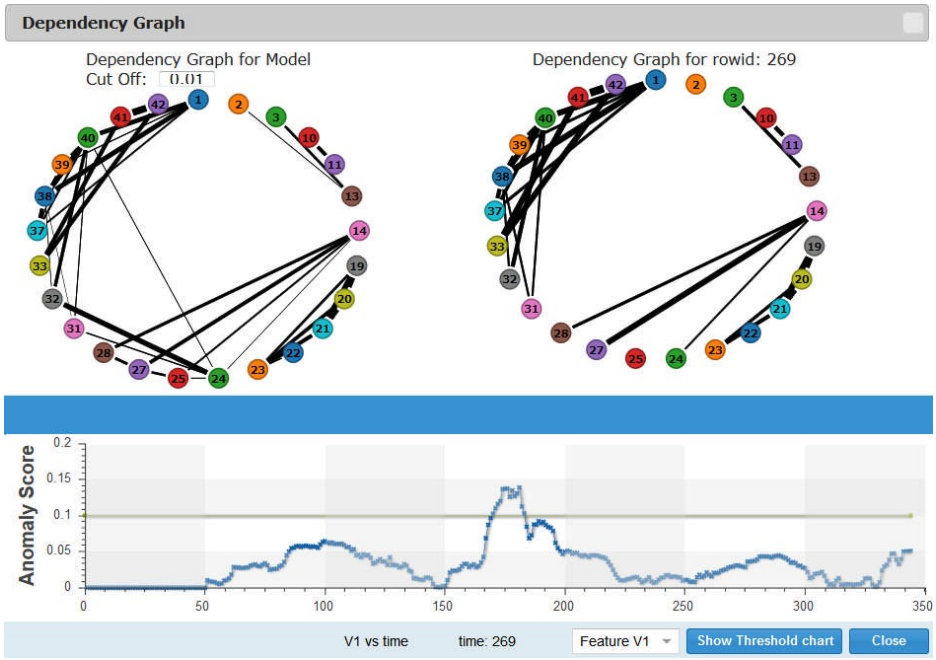

Figure 11: A Sample Output from a Failure Risk Analytics

6.5 Predictive Statistical Quality/Process Control

Predictive Statistical Quality/Process Control is a data mining approach for developing a model that predicts thequalityofprocessoutputsbasedonvariousprocessvariablesandusingthepredictiontoenhance feedback control.Typical methodsinclude ridge regress, lasso regression,customizedpredictionmodel, EWMA, causal modeling, neural networks and hierarchical modeling.

<u>Use Case: Virtual Metrology of Semiconductor Manufacturing</u>
Thisapplicationisamodelbasedpredictioncapabilityofprocessoutcomewhenthereisnophysical measurementofthatoutcome.Theunderlyingmodelsarelearnedfromhistoriesoftheactualphysical outcomesandprocesstracedata.Thetoolhelpsindetectingfaultyproductearlyandimprovingprocess control,andreducephysicalmeasurementsforprocessmonitoringandcontrol.Thisusecaseisfor prediction of plasma oxide deposition rate of a semiconductor fabrication process.

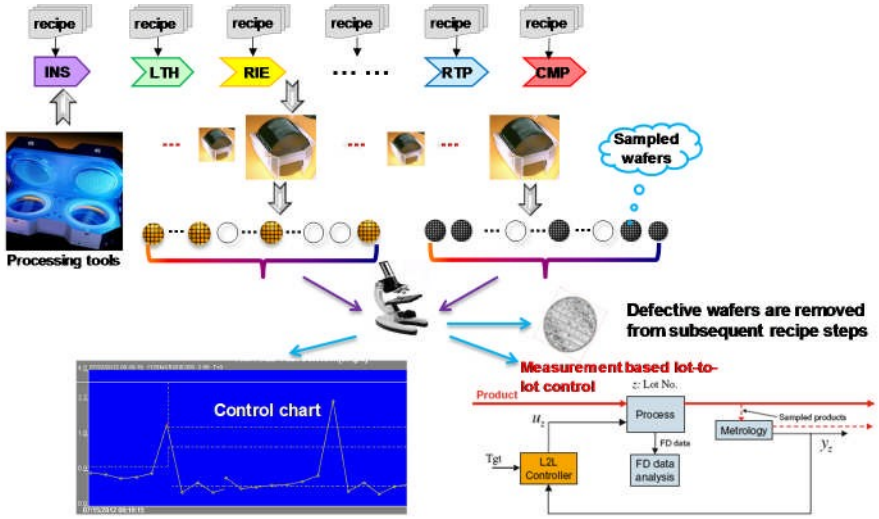

Figure 12: An Illustration of a Predictive Statistical Quality/Process Control Analytics

6.6 Model Predictive Control

ModelPredictiveControl(MPC)isanoptimizationmethodofprocesscontrolthataretypicallyused continuousprocessessuchaschemicalandpetroleumplantsandpowersystems.MPCusesdynamic modelsoftheprocess,typicallyphysics-basedmodelsordata-drivenmodelthatarecalibratedusing dynamic sensor data.The optimal control problem is typically solved as a Nonlinear Programming (NLP). MPC predicts future events and computes the optimal control profile accordingly.

<u>Use Case: Optimal Control of HVAC System</u>

This is a model predictive control (MPC) that optimally determines control profiles of the HVAC (Heating Ventilation and Air Conditioning) system as demand response. The thermal behavior of the building zone is modelled by a Nonlinear Autoregressive Neural Network (NARNET) and the optimal control problem isformulatedasaMixed-IntegerNon-LinearProgramming(MINLP)model.Theoptimalcontrolmodel minimizesthetotalcostsofpoweringHVACsystemandcorrespondingGHGemissionconsidering dynamic demand response signal, on-site energy storage system and on-site energy generation system while satisfying thermalcomfort ofbuilding occupantswithinthe physical limitationof HVACequipmentand physical limitation of on-site energy storage system and on-site energy generators.

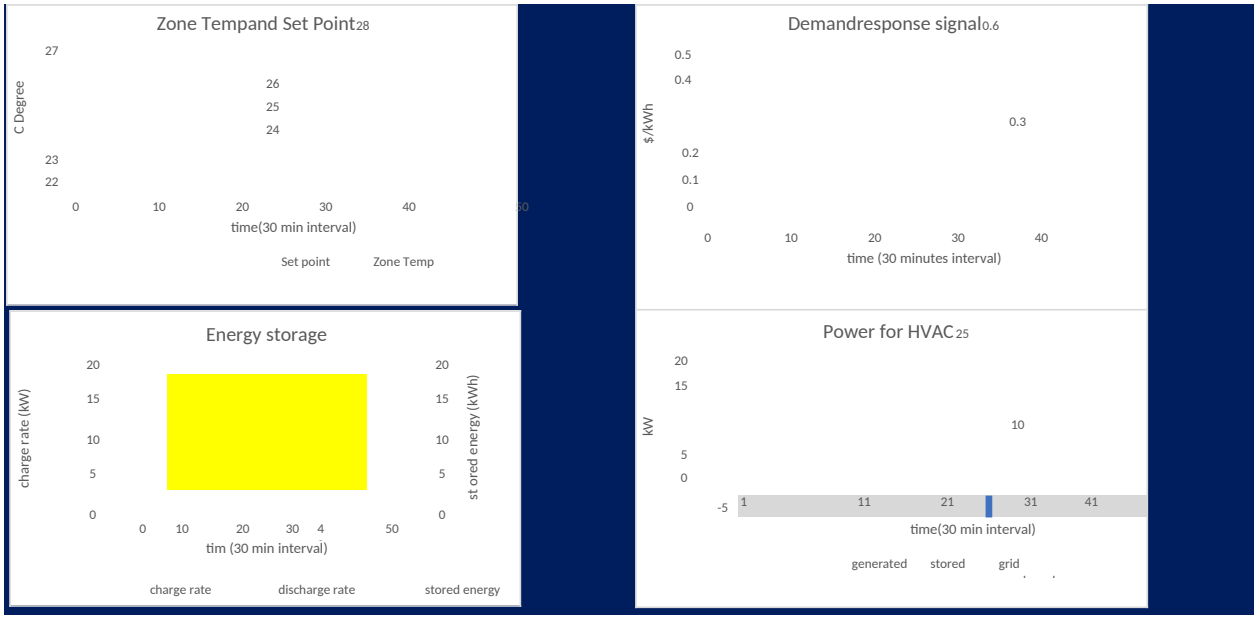

Figure 13: A Sample Output from a Model Predictive Control Analytics

6.7 Causal Modeling

Causal Modeling is a quantitative representation of cause and effect of real-world dynamics. A causal model describes the causal and other relationships, among a set of variables. Causal models incorporate the idea of multiple causality, that is, there can be more than one cause for any particular effect.

<u>Use Case: Causal Analysis of IT Operations</u>

Thistoolmonitors,detectsanddiagnosesanomaliesincomputernetworksandsystems. Withthe prevalence of IT data monitoring techniques and the surge of IT operations data available for analysis, there areincreasingexpectationsonthemanybenefitsofanalyticsbasedoperationsmanagement,suchas increasingoperationalefficienciesandunderstandingproblemsthathaveoccurred.Thisanalyticscan identify problems before they interrupt services..Additionally, it is hoped to be able to assist not only in identifying anomalous behaviors that deviate from the baseline, but in providing root cause analysis, and hence proactive failure prevention

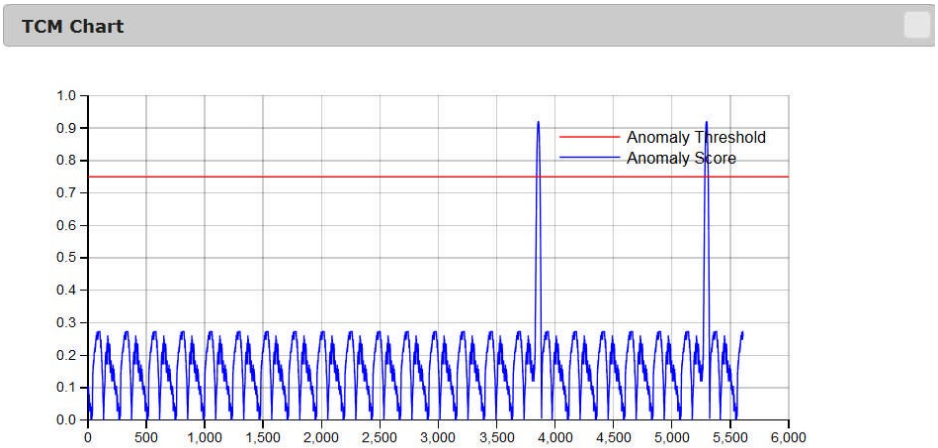

Figure 14: A Sample Output from a Causal Modeling Analytics

7 CONCLUSION

Analyticsistheengineforconvertingtheworld'snewnaturalresource,data,intogoodproductsand servicesinindustries. Withoutanalytics,thelargeamountofdatawecollectissimplydata,i.e. underutilized natural resources.By utilizing the data properly, manufacturing firms can improve product quality,reduceproductioncosts,improvesafetyinmanufacturingplants,improveenergyefficiencyof production, and improve the profit margin.Analytics can bring competitive advantage.The data can also be used to improve the quality of human lives by promoting smarter buildings, smarter cities and smarter planet.Theanalyticsintheneweraofcomputingiscomplexaggregationofvariousalgorithmsand techniques including advanced mathematics, statistics, data mining, machine learning, cognitive computing andoptimization.Therefore,ittakesalotofexpertise,effortandtimetodevelopanduseanalyticsin industries.There is an urgent need to develop and make available analytic solutions and services libraries inindustrythatcanhelpfirmsutilizetheanalyticseasilywithoutsignificantinvestmentinresources, expertiseandtime.Thecrossindustryanalyticssolution&serviceslibraryforresourceandoperations management we introduce here is intended to spark the engine.

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